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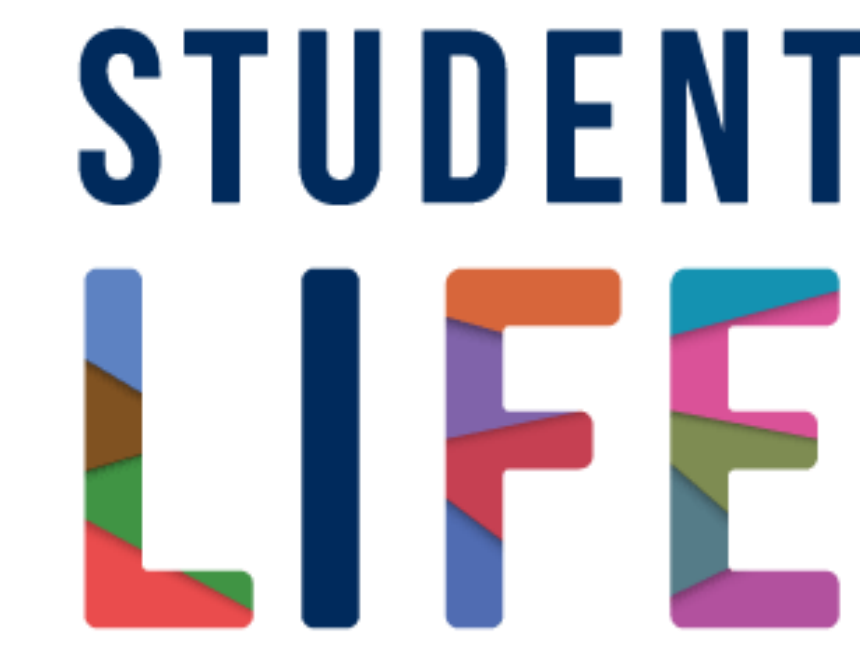
Identifying Student Profiles Through Time Allocation: A Latent Profile Analysis Approach

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Background

Abstract:

This poster presents the university student time allocation using the National Survey of Student Engagement data. Three distinct student profiles - Balanced, Academic, and Worker - were identified through latent profile analysis. This research further explores the characteristics of each profile using multinomial logistic regression, focusing on factors such as citizenship and a sense of community belonging. The poster will explain how these characteristics influence student engagement and conclude for the three distinct groups of students.

National Survey of Student Engagement (NSSE)

- All first and final year students were invited to participate
- 27% response rate (approximately 6000 responses)
- Questions included academic participation, co-curricular activities, social life, sense of belonging, and key engagement indicators
- We included the question:

About how many hours do you spend in a typical 7-day week doing the following?

- Preparing for class (studying, reading, writing, doing homework or lab work, analyzing data, rehearsing, and other academic activities)
- Participating in co-curricular activities (organizations, campus publications, student government, fraternity or sorority, intercollegiate or intramural sports, etc.)
- Working for pay on campus
- Working for pay off campus
- Doing community service or volunteer work
- Relaxing and socializing (time with friends, video games, TV or videos, keeping up with friends online, etc.)
- Providing care for dependents (children, parents, etc.)
- Commuting to campus (driving, walking, etc.)

Research Questions:

- How are students categorized by their use of time?
- How do different groups of students (based on their time allocation) differ in key engagement indicators?

Learning Outcomes

- Know how to analyze and categorize student time allocation patterns based on data from the National Survey of Student Engagement (NSSE);
- Gain insights into how different student profiles relate to various factors such as a sense of community, on-campus involvement, and demographic characteristics;
- Understand how NSSE data can inform institutional student support and well-being decisions.
- Be able to interpret the results of statistical models, such as LPA and MLR, in the context of student engagement research.

Method

Analysis Step 1:

Latent Profile Analysis (LPA)

- Statistical modelling approach that uncovers hidden subgroups within dataset by categorizing individuals based on their similarities on key factors
- Model comparison (compared between 2 to 6 group models)
- Use model fit indices to determine the most suitable number of groups or “profiles”

Analysis Step 2:

Multinomial Logistic Regression (MLR)

- Used to determine the factors that influence a student’s probability of belonging in one of the profiles
- Key factors include: Age, Sex, Citizenship, and Indicators of Sense of Community Belonging

Results

Step 1: LPA – Determining the Time Allocation Student Profiles

Class	AIC	BIC	Adjusted BIC	Entropy
2	27205.18	27613.05	27893.16	0.4477316
3	27104.90	27719.75	28142.00	0.4045657
4	27118.98	27940.81	28505.20	0.7684206
5	27137.53	28166.34	28872.87	0.8496249
6	27073.88	28309.68	29158.35	0.6037317

As we want to observe distinct groups of students, we chose the model with three classes rather than two classes. Besides, the model with three classes has a relatively smaller BIC and adjusted BIC.

Graph: Average Weekly Hours Distributed Across Activities by Three Profiles

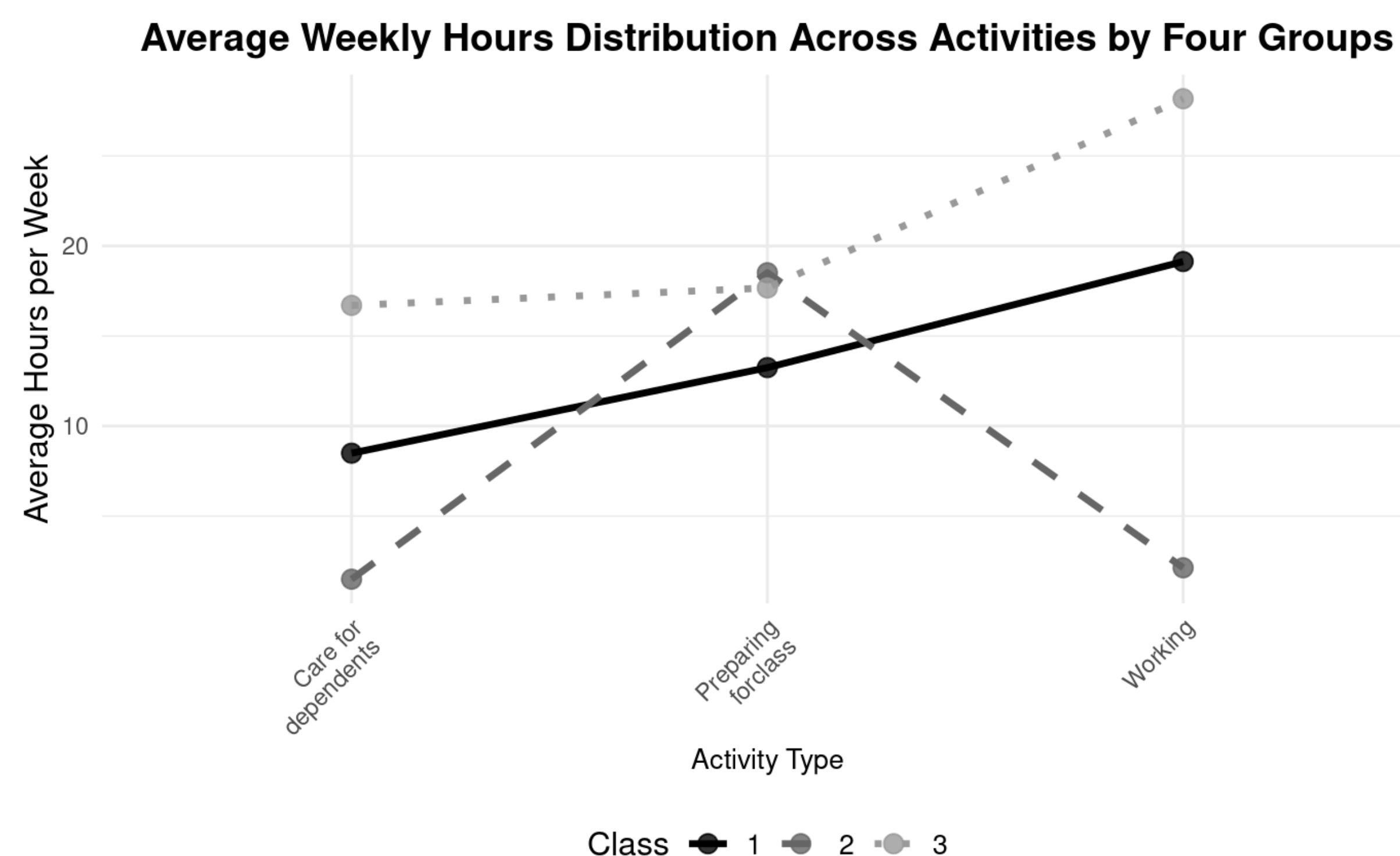


Table: Average Weekly Hours Distributed Across Activities by Three Profiles

Activity	Profile 1: Balanced 6%	Profile 2: Academic 89%	Profile 3: Worker 5%
Caring for Dependents	8.50	1.50	16.70
Preparing for Class	13.25	18.52	17.67
Working	19.14	2.13	28.18

Step 2: MLR – The probability for each factor endorsed by Academic/Worker compared to the Balanced Profile:

Factors	Academic (RRR)	Worker (RRR)
Male	1.19	1.45
International	1.46	2.27**
Sense of Belonging: Where do you find your community?		
Residence	3.12***	1.52
Student organizations	1.14	0.97
On-campus job	0.19***	0.72
Learning community	0.90	1.50
Athletics/recreation programs	0.80	0.78
Study group	1.15	1.66*
In a class	1.70**	0.86
In my academic program	1.13	0.99
Student activity space	0.86	0.44*
Religious or faith-related organization	1.43	1.19
Did not experience sense of community	1.40	0.48
Age	0.84***	0.87

1. Significance:

*p < 0.05; **p < 0.01; ***p < 0.001

2. RRR: Relative Risk Ratio

3. Base model is the Balanced Class

4. Age is a covariate